Akoizo Original Practice - Form 5 - Answer key

Division B | Elastic Launched Gliders | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which material is banned?

Unit: Construction: light, stiff, and legal.
Worked-case data: A glider weighs 3.4 g and its CG is behind the target. What should the team do?
Separate investigation observations:
Flights 1-4: 22-24 s each.
Flight 5: Hard landing on nose.
Flights 6-8: 9-12 s; glider veers right at launch.

A. Boron

B. Balsa

C. Carbon fiber

D. Plastic film

Correct: A

2. (2 points) Why does a real glider climb lower than E ÷ (mg)?

Unit: Launch energy: from stretched elastic to height.
Worked-case data: An elastic with k = 20 N/m is stretched 0.30 m. Half the energy reaches a 6.0 g glider. Find launch speed and ideal height.
Separate investigation observations:
Pull 25 cm: Peak 6 m; flight 19 s.
Pull 35 cm: Peak 9 m; flight 26 s.
Pull 45 cm: Hit 10 m ceiling; flight 15 s.

A. Drag and energy left in the elastic reduce the energy for height

B. Gravity is weaker indoors

C. Mass increases

D. Lift pushes it down

Correct: A

3. (2 points) What happens if the elastic flies off the handle during a launch?

A. Timing ends for that flight

B. The flight is repeated

C. Nothing

D. The team is disqualified

Correct: A

4. (4 points) How should a team adjust for a lower ceiling?

Unit: Launch technique and the ceiling.
Worked-case data: In practice, a 35 cm pull reaches 9 m. Competition ceiling is 7.5 m. Estimate a starting pull, assuming height scales with energy.
Separate investigation observations:
Practice room: 12 m ceiling; best pull 40 cm.
Competition room: 6 m ceiling with beams at 5.5 m.
First trim flight: 40 cm pull; glider hit a beam.

answer: Reduce the pull distance or add bank so the glider peaks just below the ceiling. (4 points).

5. (4 points) Why does decalage cause a nose-up moment at launch speed?

Unit: Trim: the transition from climb to glide.
Worked-case data: A glider climbs well but loops over at the top and then dives. Suggest a fix.
Separate investigation observations:
Launch at 0° bank: Glider loops over and dives; 8 s.
Launch at 30° bank: Climbs in a curve, rolls level; 22 s.
Launch at 70° bank: Climbs low and spirals; 12 s.

answer: The pitching moment from decalage grows with speed squared, so it is much larger in the fast climb. (4 points).

6. (4 points) Why might a heavier glider score better despite gliding faster?

Unit: Glide physics: sink rate and glide ratio.
Worked-case data: A glider reaches 9.0 m and glides at 3.2 m/s with glide ratio 8. Estimate glide time.
Separate investigation observations:
Glider A: Peak 10 m; sink 0.55 m/s.
Glider B: Peak 8 m; sink 0.35 m/s.
Question: Which glides longer, ignoring climb time?

answer: It can store more launch energy and climb higher, giving more height to glide from. (4 points).

7. (4 points) Why record peak height with flight time?

Unit: The flight log and competition day.
Worked-case data: A team has five official flights in 6 minutes. After a trim flight, it plans official flights. Its first three are 21, 23, and 8 s (a beam hit). How should it use the remaining two?
Separate investigation observations:
Flight 4: Peak 9 m, time 12 s.
Flight 5: Peak 4 m, time 13 s.
Flight 6: Peak 9 m, time 24 s after reducing rudder offset.

answer: It separates launch problems from glide problems: low peaks mean launch issues, short glides from good peaks mean trim issues. (4 points).

8. (4 points) What should be checked after a hard landing?

Unit: Construction: light, stiff, and legal.
Worked-case data: A glider weighs 3.4 g and its CG is behind the target. What should the team do?
Separate investigation observations:
Flights 1-4: 22-24 s each.
Flight 5: Hard landing on nose.
Flights 6-8: 9-12 s; glider veers right at launch.

answer: Wing and tail positions, warps, fuselage straightness, the elastic hook, and secured ballast. (4 points).

9. (4 points) How can a team make each launch use the same energy?

Unit: Launch energy: from stretched elastic to height.
Worked-case data: An elastic with k = 20 N/m is stretched 0.30 m. Half the energy reaches a 6.0 g glider. Find launch speed and ideal height.
Separate investigation observations:
Pull 25 cm: Peak 6 m; flight 19 s.
Pull 35 cm: Peak 9 m; flight 26 s.
Pull 45 cm: Hit 10 m ceiling; flight 15 s.

answer: Pull the elastic the same distance each time using marks or a fixed arm position, with the same elastic. (4 points).

10. (6 points) Solve this case and explain your reasoning: A team with a complete log flies official flights of 22.4, 18.0, 25.1, 9.6, and 24.0 s. What is the final score and tiebreaker?

reasoning: Three best times: 25.1 + 24.0 + 22.4 = 71.5 s.
Final score = 71.5 x 1.2 = 85.80.
Tiebreaker: the longest non-scored flight, 18.0 x 1.2 = 21.60. (3 points).

conclusion: Final score 85.80, tiebreaker 21.60. (3 points).

11. (2 points) What should the team change?

Supplied synthetic observations:
Practice room: 12 m ceiling; best pull 40 cm.
Competition room: 6 m ceiling with beams at 5.5 m.
First trim flight: 40 cm pull; glider hit a beam.

A. Use a thinner wing

B. Reduce pull and launch away from the beams

C. Pull farther

Correct: B

12. (2 points) What explains the best result?

Supplied synthetic observations:
Launch at 0° bank: Glider loops over and dives; 8 s.
Launch at 30° bank: Climbs in a curve, rolls level; 22 s.
Launch at 70° bank: Climbs low and spirals; 12 s.

A. Moderate bank turns the climb without wasting height

B. Bank always reduces flight time

C. Steeper bank always climbs higher

Correct: A

13. (2 points) Define “Glide ratio” in the context of Elastic Launched Gliders.

answer: Horizontal distance per unit height lost; equals L/D. (2 points).

14. (2 points) Define “Peak flight height” in the context of Elastic Launched Gliders.

answer: Highest point after launch; a required log parameter. (2 points).

15. (2 points) Define “Elastic hook” in the context of Elastic Launched Gliders.

answer: Notch or hook where the elastic engages. (2 points).

16. (2 points) Define “Kinetic energy” in the context of Elastic Launched Gliders.

answer: ½mv^2, energy of motion. (2 points).

17. (2 points) Define “Launch handle” in the context of Elastic Launched Gliders.

answer: Under 1 m long, held by one student, elastic attached. (2 points).

18. (2 points) Define “Launch routine” in the context of Elastic Launched Gliders.

answer: Rehearsed sequence of steps for each launch. (2 points).